

- Q.26 Explain the basic instrumentation system with neat block diagram.
- Q.27 What are the advantages and disadvantages of X-Y chart recorder?
- Q.28 Grounding of an instrument is necessary. Why?
- Q.29 Define following terms in op-amp
- Differential voltage gain
 - Common Mode Rejection Ratio
- Q.30 What is an ideal operational amplifier? Explain its all characteristics.
- Q.31 Write a short note on V-I characteristics of LED.
- Q.32 What are the static characteristics of an instrument?
- Q.33 Write about the working theory of liquid crystal display.
- Q.34 How an Op-Amp can be used as an instrumentation amplifier?
- Q.35 What is a printing device and state some advantages of printing devices?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What is error? Explain all types of error.
- Q.37 Explain the working principle of strip chart recorder with its constructional details. Write its advantages over X-Y chart record.
- Q.38 Derive the expression for closed loop gain of an inverting amplifier. Also write some applications of it.

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3rd Sem / Instrumentation & Control Engineering, EI Sub : Basics of Instrumentation/Pr. of Inst.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 _____ is a consistent proportional difference between the observed and true values of something.
- Gross error
 - Instrument error
 - Random error
 - Systematic error
- Q.2 The instrument which gives us directly the magnitude of physical quantity to be measured.
- Absolute
 - Indicating
 - Secondary
 - None
- Q.3 The _____ is the smallest, detectable change in the input signal that it can reliably measure and display.
- Resolution
 - Sensitivity
 - Hysteresis
 - None of the above
- Q.4 LED stands for
- Light emitting display
 - Liquid emitting display
 - Liquid emitting diode
 - Light emitting diode

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- Q.5 Strip chart recorder is a
- Measuring Instrument
 - Voltage source
 - Current divider
 - Musical tool
- Q.6 The slew rate of an ideal op-amp should be
- Very high
 - Zero
 - Infinite
 - Very low
- Q.7 Which of the following is dynamic characteristic of an instrument?
- Resolution
 - Sensitivity
 - Loading effect
 - Time constant
- Q.8 Negative feedback in op-amp
- Increases the input and output impedances
 - Increases the input impedance and bandwidth
 - Increases the output impedance and bandwidth
 - Does not affect impedance and bandwidth
- Q.9 LEDs operates at:
- Forward bias
 - Reverse bias
 - Neutral
 - None of the above
- Q.10 _____ is the systematic, automated process of recording and storing measurements such as temperature, pressure, voltage, or current over time using sensors and micro processors.
- Strip chart
 - Data logging
 - Scanning
 - All of the above

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define drift.
- Q.12 What is the use of printing device?
- Q.13 Define gross error.
- Q.14 _____ the time required for the output to reach 63.2% of its final value following a step change.
- Q.15 Expand LCD.
- Q.16 Expand GPIB.
- Q.17 The resolution is the smallest, detectable change in the input signal that it can reliably measure and display. (True/False)
- Q.18 Full form of LED.
- Q.19 What is input offset current for an ideal op-amp?
- Q.20 Define calibration.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on dynamic characteristics of an instrument.
- Q.22 Describe seven segment display in detail.
- Q.23 What is the role of instruments in any measurement process?
- Q.24 Explain accuracy and precision with suitable example.
- Q.25 What are the types of errors in measurement?

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